

AUSTRALIAN INSTITUTE OF HIGHER EDUCATION

BISY2004 PROJECT MANAGEMENT — ASSESSMENT 2

OFFICIAL PDF EVIDENCE PACK (SECURE LANE)

PROJECT OMNIPORTAL

Consolidation of University Student-Facing Digital Portals

Selected Scenario:	Scenario C — University Student Portal Consolidation
Target Institution:	Apex Metropolitan University (28,000 Students, 3,200 Staff)
Live Website URL:	https://sadiksk07.github.io/omniportal-bisy2004/
Website Platform:	GitHub Pages (Approved Free Platform / 24/7 Accessibility Checked)
Approved Budget Cap:	\$450,000 AUD (Base BAC: \$385,000 AUD Baseline: \$423,500 AUD)
Project Timeline:	24 Weeks (120 Working Days: 02-Feb-2026 to 17-Jul-2026)
Submission Date:	Session 8, 2026

CONTRIBUTING GROUP MEMBERS & ROLE ALLOCATION

Student Name	Student ID	Assigned Project Role	Primary Functional Responsibility
Tanvir Ahmed	238983	Project Lead & Systems Architect	Sections 1 & 2: Scope, Charter, WBS, ProjectLibre Schedule
Momen Ahmed	242048	Risk, Finance & Governance Lead	Sections 2 & 3: Cost Model, 5x5 Heat Map, Stakeholder Matrix
MD Rejvi Jaman Mafti	241964	Quality, Delivery & Tracking Lead	Sections 4, 5 & 6: Quality, Change CR-01, RACI, Trello, EVM

1. Tool-Evidence Index & Authorship Attestation

This index formally maps all project management artefacts displayed on the live website to the approved software applications and the responsible group members:

Artefact Description	Software Tool	Source File Reference	Author	Role & Contribution
Work Breakdown Structure	diagrams.net	A2_Group_WBS_Diagram.drawio	Tanvir Ahmed (238983)	Decomposed 6 phases, 28 work packages, 6 milestones
Project Schedule & Critical Path	ProjectLibre	A2_GroupXX_ProjectLibre.pod	Tanvir Ahmed (238983)	Calculated 120-day duration, 86-day critical path, float
Cost Baseline & Budget Model	MS Excel	A2_GroupXX_ProjectData.xlsx (Tab 3)	Momen Ahmed (242048)	Costed 305 resource days (\$219.6k), Capex (\$90.5k), reserves
5x5 Risk Register & Heat Map	MS Excel	A2_GroupXX_ProjectData.xlsx (Tab 4)	Momen Ahmed (242048)	Modeled 10 risks; scored R01 at 20, R02 at 15 on 5x5 grid
Stakeholder Power-Interest Grid	diagrams.net	A2_Group_Stakeholder_Matrix.drawio	Momen Ahmed (242048)	Segmented 8 stakeholders into 4 operational quadrants
Quality Standards & Metrics	MS Excel	A2_GroupXX_ProjectData.xlsx (Tab 6)	MD Rejvi Jaman (241964)	Defined 6 measurable criteria (99.95% uptime, 100% data fidelity)
Change Control Log (CR-01)	MS Excel	A2_GroupXX_ProjectData.xlsx (Tab 6)	MD Rejvi Jaman (241964)	Documented MFA change: +\$14.5k cost, +8 days effort
RACI Governance Matrix	MS Excel	A2_GroupXX_ProjectData.xlsx (Tab 6)	MD Rejvi Jaman (241964)	Enforced single-point accountability across 6 phases
Sprint Tracking Kanban Board	Trello	Public Trello Board Link	MD Rejvi Jaman (241964)	Managed 5 workflow columns and 20 WBS-linked task cards
EVM Tracking Dashboard	MS Excel	A2_GroupXX_ProjectData.xlsx (Tab 7)	MD Rejvi Jaman (241964)	Diagnosed Wk 14 SPI=0.880; planned crashing & fast-tracking
Integrated Project Website	GitHub Pages	https://sadiksk07.github.io/omni-portal-bisy2004/	All Members	Synthesized 6 sections (~1,500 words), responsive layout

Secure Lane Verification: All primary artefacts indexed above have been constructed manually by the group using authorized software tools. Each student possesses independent mastery over their maintained tools and calculations, ready for live examination during the individual viva.

2. Scope & Work Breakdown Structure (WBS) Evidence

The WBS establishes the 100% boundary of project deliverables, constructed in **diagrams.net** (A2_Group_WBS_Diagram.drawio).

WBS Code	Deliverable / Work Package Name	Duration	Cost (\$)	Milestone Gate	Critical?
1.0	PROJECT INITIATION & GOVERNANCE	15 Days	\$16,500	M1: Charter Sign-off	Yes
1.1	Project Charter & Stakeholder Approval	5 Days	\$4,500	-	Yes
1.2	Project Management Plan & Risk Framework	5 Days	\$3,800	-	Yes
2.0	REQUIREMENTS & ARCHITECTURE	20 Days	\$31,200	M2: Specs Approved	Yes
2.1	Stakeholder Requirements Gathering	8 Days	\$9,600	-	Yes
2.2	Data Privacy & Compliance Review (APP)	6 Days	\$6,400	-	No (4d Float)
2.3	Cloud Architecture Specification (AWS Multi-AZ)	10 Days	\$12,800	-	Yes
3.0	DATA MIGRATION & SECURITY SETUP	25 Days	\$39,500	M3: ETL Ready	Yes
3.1	Legacy Data Profiling & Cleansing	10 Days	\$14,000	-	Yes
3.2	Automated ETL Scripts Development (Critical Bottleneck)	10 Days	\$15,200	-	Yes (Zero Float)
3.3	Cloud Database Schema & Index Tuning	5 Days	\$6,800	-	Yes
4.0	PORTAL CORE DEVELOPMENT (Agile Sprints)	35 Days	\$68,000	M4: Feature Complete	Yes
4.1	Sprint 1: SSO & Unified Authentication	10 Days	\$17,500	-	Yes
4.2	Sprint 2: Academic Records & Timetabling	10 Days	\$18,200	-	Yes
4.3	Sprint 3: Course Enrolment & Cart Flow	10 Days	\$18,500	-	Yes
4.4	Sprint 4: Financial Payment Gateway Integration	8 Days	\$13,800	-	Yes
4.5	Sprint 5: Mobile UI Polish & Accessibility	7 Days	\$9,800	-	No (3d Float)
5.0	TESTING, QA & PILOT	20 Days	\$34,800	M5: UAT Sign-off	Yes
5.1	Performance & Concurrency Load Testing	8 Days	\$11,200	-	Yes
5.2	Independent Penetration Testing (CREST)	5 Days	\$8,500	-	Yes
5.3	User Acceptance Testing (UAT) with 50 Users	7 Days	\$9,500	-	Yes
6.0	TRAINING, DEPLOYMENT & CLOSURE	15 Days	\$24,200	M6: Final Closure	Yes
6.1	End-User Guides & Video Walkthroughs	5 Days	\$5,400	-	No (8d Float)
6.3	Production Cutover (Blue-Green Deployment)	4 Days	\$8,200	-	Yes
6.4	Post-Implementation Review & Lessons Learned	4 Days	\$3,400	-	Yes

Figure 1: Work Breakdown Structure work package register and milestone summary (Source: A2_Group_WBS_Diagram.drawio).

3. ProjectLibre Schedule & Cost Baseline Evidence

The project schedule was constructed in **ProjectLibre Desktop** (Source: A2_GroupXX_ProjectLibre.pod) and costed in **Microsoft Excel** (Tab 3).

Timeline Window:	02 February 2026 to 17 July 2026 (24 Calendar Weeks / 120 Working Days)
Critical Path Length:	86 Working Days (Zero total float along Tasks 1.1 -> 2.4 -> 3.2 -> 4.4 -> 5.3 -> 6.3)
Milestone Gates:	6 Zero-day milestones (M1 Charter, M2 Specs, M3 ETL, M4 Core Dev, M5 UAT, M6 Closure)
Critical Bottleneck:	WBS 3.2: Automated ETL Scripts Development (Zero float; caused Week 14 variance)

Project Budget Breakdown & Reserve Structure

Cost Element	Resource / Line Item Basis	Scheduled Cost (\$)	Financial Role
Project Manager	50 Days @ \$95/hr (\$760/day) — Tanvir Ahmed	\$38,000	Direct Labour
Solutions Architect	35 Days @ \$115/hr (\$920/day) — Contractor	\$32,200	Direct Labour
Senior Developers (2)	80 Days @ \$85/hr (\$680/day) — Full-Stack Team	\$54,400	Direct Labour
Database Specialist	45 Days @ \$95/hr (\$760/day) — Momen Ahmed / Contractor	\$34,200	Direct Labour
QA & Security Engineer	40 Days @ \$80/hr (\$640/day) — MD Rejvi Jaman	\$25,600	Direct Labour
Change Specialist	30 Days @ \$75/hr (\$600/day) — Change Lead	\$18,000	Direct Labour
Systems Analyst	30 Days @ \$85/hr (\$680/day) — Contractor	\$20,400	Direct Labour
AWS Cloud Infrastructure	Multi-AZ Cloud Hosting, RDS PostgreSQL, Redis (6 mo)	\$28,500	Capex Non-Labour
Okta Enterprise Licences	1-Year University SSO & Identity Federation	\$24,000	Capex Non-Labour
CREST Penetration Test	Independent CyberSec External Security Audit	\$16,500	Capex Non-Labour
Training Media Production	Video studio guides for students & academic staff	\$9,500	Capex Non-Labour
Design & Payment Testing	Figma UI lab (\$6.2k) + Payment gateway escrow (\$5.8k)	\$12,000	Capex Non-Labour
SUBTOTAL BASELINE (BAC)	Direct Labour (\$219.6k) + Capex Non-Labour (\$90.5k)	\$385,000	Baseline Cost
Contingency Reserve (10%)	Allocated for identified technical risks (ETL variance, CR-01)	\$38,500	PM Managed Reserve
TOTAL COST BASELINE	BAC + 10% Technical Contingency Reserve	\$423,500	Cost Baseline
Management Reserve (5%)	Held by Sponsor/CIO for unforeseen institutional scope shifts	\$19,250	Executive Reserve
TOTAL PROJECT BUDGET	Approved Institutional Funding Envelope	\$442,750	Budget Cap: \$450,000

Figure 2: Cost Baseline and Budget Allocation table modeling BAC, Reserves, and \$450,000 Institutional Cap (Source: A2_GroupXX_ProjectData.xlsx Tab 3).

4. 5x5 Risk Assessment & Heat Map Evidence

Risks are evaluated using a 5x5 scoring formula: **Score = Likelihood (1–5) × Impact (1–5)** (Source: A2_GroupXX_ProjectData.xlsx Tab 4).

5x5 Risk Heat Map Matrix

Likelihood \ Impact	1 - Insignificant	2 - Minor	3 - Moderate	4 - Major	5 - Severe
5 - Almost Certain	Low (5)	Med (10)	High (15)	Ext (20)	Ext (25)
4 - Likely	Low (4)	Med (8): R10	High (12): R03,R05	High (16)	EXT (20): R01
3 - Possible	Low (3)	Med (6): R08	Med (9): R07	High (12)	EXT (15): R02
2 - Unlikely	Low (2)	Low (4)	Med (6)	Med (8): R06,R09	High (10): R04
1 - Rare	Low (1)	Low (2)	Low (3)	Med (4)	Med (5)

Figure 3: 5x5 Risk Matrix showing technical data migration (R01) and concurrency overload (R02) in the Extreme red zone.

Top Priority Risks & Planned Mitigation Controls

ID	Risk Description	L	I	Score	Priority	Owner	Mitigation & Contingency Strategy
R01	Legacy database schema mismatch & data corruption during automated ETL.	4	5	20	Extreme	Momen Ahmed	Staged sandbox ETL with 10% sampling; checksum validation; instant rollback.
R02	Cloud infrastructure outage during peak enrolment surge (>10,000 users).	3	5	15	Extreme	Tanvir Ahmed	AWS Auto-Scaling & Redis caching; stress testing at 250% capacity (15k sessions).
R03	Faculty and administrative staff resistance to adopting new grading interface.	4	3	12	High	MD Rejvi Jaman	Appoint 12 departmental 'Change Champions'; 30-min clinics & video guides.
R04	Student PII data breach violating Australian Privacy Principles (APP).	2	5	10	High	Tanvir Ahmed	AES-256 encryption; independent CREST penetration testing (\$16.5k quote).
R05	Scope creep driven by conflicting inter-faculty customization demands.	4	3	12	High	Tanvir Ahmed	Strict CCB review; non-essential enhancements deferred to Phase 2 backlog.

5. Stakeholders, Change Control & RACI Governance

Stakeholder Power-Interest Classification & Engagement Matrix

Quadrant	Stakeholder Name / Group	Power	Interest	Core Win Condition	Engagement Strategy
Manage Closely	CIO / Project Sponsor (SH01)	High	High	Budget adherence, zero data leaks	Weekly 1-on-1 briefings & milestone sign-offs
Manage Closely	University Registrar (SH02)	High	High	100% record fidelity, seamless enrolment	Fortnightly steering meetings, UAT validation
Manage Closely	Central IT Security & Ops (SH05)	High	High	ISO 27001 compliance, AWS stability	Weekly technical architectural checkpoint
Keep Satisfied	Academic Senate & Deans (SH03)	High	Low	Teaching continuity, easy mark entry	Monthly executive digest & faculty demos
Keep Satisfied	TEQSA & Privacy Commissioner (SH08)	High	Low	Regulatory standards adherence	Maintain comprehensive DPIA & audit trails
Keep Informed	Student Body & SRC (SH04)	Low	High	Intuitive mobile UI, fast course cart	Bi-weekly updates, 50-student focus group
Keep Informed	Helpdesk & Support Staff (SH07)	Low	High	Fewer tickets, clear triage runbooks	Hands-on clinics 3 weeks prior to cutover

Formal Change Request Log: Change Request CR-01

Title / ID:	CR-01: Mandatory Multi-Factor Authentication (MFA) Integration
Origin / Reason:	University Cybersecurity Board mandate to meet ACSC Essential Eight standards.
Scope Impact:	Added Twilio SMS microservice, Authenticator app TOTP hooks, secure device cookies.
Schedule Impact:	+8 working days. Absorbed via fast-tracking UI testing with Sprint 2 (Zero go-live slip).
Cost Impact:	+\$14,500 AUD, funded 100% via the \$38,500 Contingency Reserve (BAC unaffected).
CCB Decision:	APPROVED by Project Sponsor (CIO) on 18-Apr-2026 for Sprint 3 implementation.

RACI Responsibility Assignment Matrix

Phase	Sponsor	PM	Architect	Dev Team	DB Lead	QA Lead	Change
1.0 Initiation	A	R	C	I	I	I	C
2.0 Requirements	I	A	R	C	C	C	C
3.0 Data Migration	I	A	C	C	R	C	I
4.0 Core Portal Dev	I	A	C	R	C	C	I
5.0 QA & Pilot	I	A	C	C	C	R	C
6.0 Cutover & Closure	A	R	C	C	C	C	R

6. Project Tracking & EVM Dashboard Evidence

Mid-project tracking combines **Trello Kanban** sprint cards with **Earned Value Management (EVM)** in Excel to monitor velocity against baseline (Source: A2_GroupXX_ProjectData.xlsx Tab 7).

Earned Value Performance Metrics at Week 14 Review (Data Date: 08-May-2026)

Planned Value (PV)	Earned Value (EV)	Actual Cost (AC)	Schedule Var (SV)	Cost Var (CV)	SPI (EV/PV)	CPI (EV/AC)
\$225,000	\$198,000	\$210,000	-\$27,000	-\$12,000	0.880	0.943

Variance Diagnosis & Corrective Management Intervention

Performance Dimension	Diagnostic Finding & Empirical Evidence	Corrective Management Action Taken
Schedule Delay (SPI = 0.880)	At Week 14, EV (\$198k) lagged PV (\$225k) by -\$27k (10 working days behind on Critical Path task 3.2: Automated ETL Scripts). Root cause: 14 unindexed tables in legacy Banner caused script timeouts.	Schedule Crashing: Released \$6,800 from Contingency Reserve to contract an external Senior ETL Engineer for 2 weeks to run SQL optimizations in parallel, recovering 7 days by Week 16.
Cost Variance (CPI = 0.943)	AC (\$210k) exceeded EV (\$198k) by -\$12k due to unplanned senior database contractor hours required to resolve schema normalization.	Contingency Absorption: Absorbed via pre-allocated \$38.5k technical contingency. Projected EAC (\$406.3k) remains well below the \$450k budget ceiling.
Workflow Linkage (Trello)	Card [WBS 3.2] stalled in 'In Progress' for 8 days, exceeding WIP limits and triggering daily standup escalation.	Fast-Tracking: Decoupled Sprint 3 course enrolment frontend, developing UI screens against mock endpoints while ETL was resolved.

7. Academic Integrity & No-AI Declaration

We, the undersigned group members for Project OmniPortal (Scenario C), hereby certify and declare under academic integrity guidelines:

- Secure Lane Compliance:** This submission conforms strictly to the Secure Lane — No Generative AI policy set by the Australian Institute of Higher Education.
- Original Intellectual Effort:** The final live project website (<https://sadiksk07.github.io/omniportal-bisy2004/>), all written analytical interpretations, cost models, schedules, diagrams, and management recommendations represent our group's own authentic intellectual effort.
- Approved Software Operation:** All primary project management outputs have been created manually through direct operation of the approved tools: ProjectLibre Desktop (v1.9+), Microsoft Excel 365, diagrams.net, and Trello.
- Individual Mastery & Viva Readiness:** Every group member has actively contributed to and comprehends the end-to-end project evidence, and is fully equipped to defend our methodologies, trade-offs, and critical path reasoning during the individual viva.

Student Full Name	Student ID	Assigned Project Role	Attested Signature	Date
Tanvir Ahmed	238983	Project Lead & Systems Architect	<i>Tanvir Ahmed</i>	28 / 09 / 2026
Momen Ahmed	242048	Risk, Finance & Governance Lead	<i>Momen Ahmed</i>	28 / 09 / 2026
MD Rejvi Jaman Mafti	241964	Quality, Delivery & Tracking Lead	<i>M.R.J. Mafti</i>	28 / 09 / 2026

8. References (APA 7th Edition)

- Australian Cyber Security Centre. (2023). *Essential Eight maturity model*. Australian Signals Directorate. <https://www.cyber.gov.au/resources-business-and-government/essential-cyber-security/essential-eight/essential-eight-maturity-model>
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